Question 1

Prove that $d(S) = d(\overline{S})$ for any set S in $\mathbb{R}^n$.

Proof:

Since $S \subset \overline{S}$, (A) $d(S) \leq d(\overline{S})$. If X and Y are in $\overline{S}$ and $\epsilon > 0$, there are points X_1, Y_1 in S such that $|X - X_1| < \epsilon$ and $|Y - Y_1| < \epsilon$. Then

$$|X - Y| \leq |X - X_1| + |X_1 - Y_1| + |Y_1 - Y| < d(S) + 2\epsilon.$$

Hence, $d(\overline{S}) < d(S) + \epsilon$. Let $\epsilon \to 0+$ to conclude that $d(\overline{S}) \leq d(S)$, which, with (A), implies that $d(S) = d(\overline{S})$.

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Question 2

If a nonempty subset S of $\mathbb{R}^n$ is both open and closed, then $S = \mathbb{R}^n$.

Prove:

Suppose that X_0 is in S. If $S \neq \mathbb{R}^n$, there is an $X_1 \notin S$. Let

$$H = \{\tau \mid (1-t)X_0 + tX_1 \in S \text{ for } 0 \leq t < \tau\}.$$

Since S is open it contains a neighborhood of X_0, so $H \neq \emptyset$. Since X_1 is not in S, $\tau \leq 1$ for all τ in H. Let $\rho = \sup H$ and $\overline{X} = (1-\rho)X_0 + \rho X_1$; then $\overline{X}$ is a limit point of S and so in S, because S is closed. Since S is open, it contains some ϵ–neighborhood of $\overline{X}$, so $\rho + \epsilon/2$ is in H. This contradicts the definition of ρ. Hence, $S = R^n$.

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Question 3

$$S = \{(x, y, x) \mid |x| \leq 2, |y| \leq 1, |z - 2| \leq 2\}$$

Ans:

$d(S)$ is the supremum of $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

where $-2 \leq x_1, x_2 \leq 2, -1 \leq y_1, y_2 \leq 1, 0 \leq z_1, z_2 \leq 4$;

to maximize this function let, for example,
$x_1 = -2, x_2 = 2, y_1 = -1, y_2 = 1, z_1 = 0, z_2 = 4$; thus, $d(S) = \sqrt{4^2 + 2^2 + 4^2} = 6$.

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Question 4

$$f(x, y, z) = \log(1 + x + y + z). \quad \Phi = (0, 1, 0)$$

Ans:

$$h(t) = f(x + \phi_1 t, y + \phi_2 t, z + \phi_3 t)$$
$$= \log(1 + x + y + z + (\phi_1 + \phi_2 + \phi_3)t);$$
$$h'(t) = \frac{\phi_1 + \phi_2 + \phi_3}{1 + x + y + z + (\phi_1 + \phi_2 + \phi_3)t};$$

$$\frac{\partial f(X)}{\partial L} = h'(0) = \frac{\phi_1 + \phi_2 + \phi_3}{1 + x + y + z}; \text{ if } L = (0, 1, 0)$$

then

$$\frac{\partial f(X)}{\partial L} = \frac{1}{1 + x + y + z}.$$

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Question 5

If $h(u, v) = f(u^2 + v^2)$, then $vh_u - uh_v = 0$.

ANS:

$$h_u = 2uf'$$
$$h_v = 2vf'$$
$$vh_u - uh_v = 2(vu - uv)f'$$
$$vh_u - uh_v = 0.$$

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Question 6

If $h(u, v) = f(\sin u + \cos v)$, then $h_u \sin v + h_v \cos u = 0$.

Ans:

$$h_u = f' \cos u$$
$$h_v = -f' \sin v$$
$$h_u \sin v + h_v \cos u = (\sin v \cos u - \cos u \sin v) f'$$
$$h_u \sin v + h_v \cos u = 0.$$

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Question 7

If $h(u, v) = f(u/v)$, then $uh_u + vh_v = 0$.

Ans:

$$h_u = \frac{1}{v} f'$$
$$h_v = -\frac{u}{v^2} f'$$
$$uh_u + vh_v = \left(\frac{u}{v} - \frac{u}{v}\right) f'$$
$$uh_u + vh_v = 0.$$

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Question 8

If $h(u, v) = f(g(u, v), -g(u, v))$, then $dh = (f_x - f_y)\, dg$.

Ans:

$$h_u = (f_x - f_y) g_u$$
$$h_v = (f_x - f_y) g_v$$
$$dh = h_u\, du + h_v\, dv$$
$$dh = (f_x - f_y)(g_u\, du + g_v\, dv)$$
$$dh = (f_x - f_y)\, dg.$$

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Question 9

Let $h(r, \theta, z) = f(x, y, z)$, where $x = r\cos\theta$ and $y = r\sin\theta$.
Find h_r, h_θ, and h_z in terms of f_x, f_y, and f_z.

Ans:

$$h_r = f_x x_r + f_y y_r + f_z z_r$$
$$h_r = f_x \cos\theta + f_y \sin\theta$$
$$h_\theta = f_x x_\theta + f_y y_\theta + f_z z_\theta$$
$$h_\theta = f_x(-r\sin\theta) + f_y(r\cos\theta)$$
$$h_\theta = r(-f_x \sin\theta + f_y \cos\theta)$$
$$h_z = f_x x_z + f_y y_z + f_z = f_z.$$

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Question 10

Let $h(r, \theta, \phi) = f(x, y, z)$, where $x = r\sin\phi\cos\theta$, $y = r\sin\phi\sin\theta$, and $z = r\cos\phi$. Find h_r, h_θ, and h_ϕ in terms of f_x, f_y, and f_z.

ANS:

$$h_r = f_x x_r + f_y y_r + f_z z_r$$
$$h_r = f_x \sin\phi\cos\theta + f_y \sin\phi\sin\theta + f_z \cos\phi$$
$$h_\theta = f_x x_\theta + f_y y_\theta + f_z z_\theta$$
$$h_\theta = r\sin\phi(-f_x \sin\theta + f_y \cos\theta)$$
$$h_\phi = f_x x_\phi + f_y y_\phi + f_z z_\phi$$
$$h_\phi = r(f_x \cos\phi\cos\theta + f_y \cos\phi\sin\theta - f_z \sin\phi.$$

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Question 11

Let $h(\mathrm{U}) = f(\mathrm{G}(\mathrm{U}))$ and find $d_{\mathrm{U}_0} h$ by writing h explicitly as a function of U.

$f(x, y) = 3x^2 + 4xy^2 + 3x.$
$g_1(u, v) = ve^{u+v-1}.$
$g_2(u, v) = e^{-u+v-1}.$
$(u_0, v_0) = (0, 1)$

Ans:

$$h(\mathrm{U}) = 3v^2e^{2u+2v-2} + 4ve^{u+v-1}e^{-2u+2v-2} + 3ve^{u+v-1}$$
$$= 3v^2e^{2u+2v-2} + 4ve^{-u+3v-3} + 3ve^{u+v-1}$$
$$h_u(\mathrm{U}) = 6v^2e^{2u+2v-2} - 4ve^{-u+3v-3} + 3ve^{u+v-1}$$
$$h_u(\mathrm{U}_0) = 5$$
$$h_v(\mathrm{U}) = (6v + 6v^2)e^{2u+2v-2} + (4 + 12v)e^{-u+3v-3} + (3 + 3v)e^{u+v-1}$$
$$h_v(\mathrm{U}_0) = 34$$
$$d_{\mathrm{U}_0} h = h_u(\mathrm{U}_0)\, du + h_v(\mathrm{U}_0)\, dv = 5\, du + 34\, dv$$

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Question 12

Let $h(\mathrm{U}) = f(\mathrm{G}(\mathrm{U}))$ and find $d_{\mathrm{U}_0} h$ by writing h explicitly as a function of U.

$f(x, y, z) = e^{-(x+y+z)}.$
$g_1(u, v, w) = \log u - \log v + \log w.$
$g_2(u, v, w) = -2\log u - 3\log w.$
$g_3(u, v, w) = \log u + \log v + 2\log w.$
$(u_0, v_0, w_0) = (1, 1, 1)$

Ans:

$$x(\mathrm{U}) + y(\mathrm{U}) + z(\mathrm{U}) = (\log u - \log v + \log w) + (-2\log u - 3\log w) + (\log u + \log v + 2\log w)$$
$$x(\mathrm{U}) + y(\mathrm{U}) + z(\mathrm{U}) = 0$$

hence

$$h(\mathrm{U}) \equiv 1$$
$$d_{\mathrm{U}_0} h = 0.$$

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Question 13

Let $h(\mathrm{U}) = f(\mathrm{G}(\mathrm{U}))$ and find $d_{\mathrm{U}_0}h$ by writing h explicitly as a function of U.

$f(x,y) = (x+y)^2$.
$g_1(u,v) = u\cos v$. $(u_0, v_0) = (3, \pi/2)$
$g_2(u,v) = u\sin v$.

Ans:

$$h(\mathrm{U}) = u^2(\cos v + \sin v)^2$$
$$= u^2(1 + 2\sin v\cos v)$$
$$= u^2(1+\sin 2v)$$
$$h_u(\mathrm{U}) = 2u(1+\sin 2v)$$
$$h_u(\mathrm{U}_0) = 6$$
$$h_v(\mathrm{U}) = 2u^2\cos 2v;$$
$$h_v(\mathrm{U}_0) = -18$$
$$d_{\mathrm{U}_0}h = h_u(\mathrm{U}_0)\,du + h_v(\mathrm{U}_0)\,dv$$
$$d_{\mathrm{U}_0}h = 6\,du - 18\,dv.$$

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Question 14

Let $h(\mathrm{U}) = f(\mathrm{G}(\mathrm{U}))$ and find $d_{\mathrm{U}_0}h$ by writing h explicitly as a function of U.

$f(x,y,z) = x^2+y^2+z^2$.
$g_1(u,v,w) = u\cos v\sin w$.
$g_2(u,v,w) = u\cos v\cos w$. $(u_0, v_0, w_0) = (4, \pi/3, \pi/6)$
$g_3(u,v,w) = u\sin v$;

Ans:

$$h(\mathrm{U}) = u^2\cos^2 v\sin^2 w + u^2\cos^2 v\cos^2 w + u^2\sin^2 v$$
$$h(\mathrm{U}) = u^2$$
$$h_u(\mathrm{U}) = 2u$$
$$h_v(\mathrm{U}) = 0$$
$$h_w(\mathrm{U}) = 0$$
$$h_u(\mathrm{U}_0) = 8;$$
$$h_v(\mathrm{U}_0) = 0;$$
$$h_w(\mathrm{U}_0) = 0$$
$$dh = 8\,du.$$

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Question 15

Show that $(0,0)$ is a critical point of each of the following functions, and that they have positive semidefinite second differentials at $(0,0)$.

$$p(x,y) = x^2 - 2xy + y^2 + x^4 + y^4;$$

Ans:

$$p(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$$
$$p_x(x,y) = 2x - 2y + 4x^3$$
$$p_y(x,y) = -2x + 2y + 4y^3.$$

Since $p_x(0,0) = p_y(0,0) = 0$, $(0,0)$ is a critical point of p.

$$p_{xx}(x,y) = 2 + 12x^2$$
$$p_{xy}(x,y) = -2;$$
$$p_{yy}(x,y) = 2 + 12y^2$$
$$(d^{(2)}_{(0,0)}p)(x,y) = 2x^2 - 4xy + 2y^2$$
$$(d^{(2)}_{(0,0)}p)(x,y) = 2(x-y)^2 \text{ is positive semidefinite.}$$

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Question 15

Show that $(0,0)$ is a critical point of each of the following functions, and that they have positive semidefinite second differentials at $(0,0)$.

$$q(x,y) = x^2 - 2xy + y^2 - x^4 - y^4.$$

Ans:

$$q(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$$
$$q_x(x,y) = 2x - 2y - 4x^3$$
$$q_y(x,y) = -2x + 2y - 4y^3$$

Since $q_x(0,0) = q_y(0,0) = 0$, $(0,0)$ is a critical point of q.

$$q_{xx}(x,y) = 2 - 12x^2$$
$$q_{xy}(x,y) = -2$$
$$q_{yy}(x,y) = 2 - 12y^2$$
$$(d^{(2)}_{(0,0)}q)(x,y) = 2x^2 - 4xy + 2y^2$$
$$(d^{(2)}_{(0,0)}q)(x,y) = 2(x-y)^2 \text{ is positive semidefinite.}$$

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Question 17

If R is degenerate, then implies that $\int_R f(\mathrm{X})\,d\mathrm{X} = 0$ if f is bounded on R.

Proof:

Suppose, for example, that $R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n]$ with $a_1 = b_1$.

Then every partition P of R is of the form $\mathrm{P} = \{R_1, R_2, \ldots, R_k\}$, where $v(R_j) = 0$, $1 \le j \le k$. Therefore, every Riemann sum of f over R equals zero, so $\int_R f(\mathrm{X})d\mathrm{X} = 0$.

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Question 18

Suppose that

$$f(x,y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational,} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational,} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational,} \\ 3 & \text{if } x \text{ and } y \text{ are irrational.} \end{cases}$$

Find $\overline{\int_R} f(x,y)\,d(x,y)$ and $\underline{\int_R} f(x,y)\,d(x,y)$ if $R = [a,b] \times [c,d]$.

Ans:

Let $\mathrm{P} = \{R_1, R_2, \ldots, R_k\}$. Let

$$M_j = \sup\{f(x,y) \mid \mathrm{X} \in R_j\} \quad \text{and} \quad m_j = \inf\{f(x,y) \mid \mathrm{X} \in R_j\}.$$

Let j be arbitrary in $\{1, 2, \ldots, k\}$. Since R_j contains a point $(\hat{x}_j, \hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational, $M_j = 3$. Hence, $\overline{\int_R} f(x,y)\,d(x,y) = 3(b-a)(d-c)$. Since R_j contains a point $(\tilde{x}_j, \tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational, $m_j = 0$. Hence, $\underline{\int_R} f(x,y)\,d(x,y) = 0$.

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Question 19

If S_1 and S_2 have zero content, then $S_1 \cup S_2$ has zero content.

Prove

If $\epsilon > 0$, there are rectangles $T_1, T_2, \ldots, T_r$ and $T_1', T_2', \ldots, T_s'$ such that $S_1 \subset \cup_{i=1}^{r} T_i$, $S_2 \subset \cup_{j=1}^{s} T_j'$, $\sum_{i=1}^{r} V(T_r) < \epsilon/2$, and $\sum_{j=1}^{s} V(T_j') < \epsilon/2$. Then $\{T_1, \ldots, T_m, T_1', \ldots, T_r'\}$ covers $S_1 \cup S_2$ with total content $< \epsilon$.

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Question 20

If S_1 has zero content and $S_2 \subset S_1$, then S_2 has zero content.

prove

If $\epsilon > 0$, there are rectangles $T_1, T_2, \ldots, T_r$ and $T_1', T_2', \ldots, T_s'$ such that If $S_1 \subset \cup_{i=1}^{r} T_r$ and $S_2 \subset S_1$, then $S_2 \subset \cup_{i=1}^{r} T_r$.

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Question 21

If S has zero content, then $\overline{S}$ has zero content.

Prove

If $\epsilon > 0$, there are rectangles $T_1, T_2, \ldots, T_r$ and $T_1', T_2', \ldots, T_s'$ such that

Since $\cup_{i=1}^{r} T_i$ is closed, $\overline{S} \subset \cup_{i=1}^{r} T_i$ if $S \subset \cup_{i=1}^{r} T_i$.

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Question 22

Show that a degenerate rectangle has zero content.

Prve:

Let $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$ and

$T = [a_1 - \epsilon, b_1 + \epsilon] \times \cdots \times [a_n - \epsilon, b_n + \epsilon]$, where $\epsilon > 0$.

Then $R \subset T$ and, since $a_i = b_i$ for some i, $V(T) \leq 2\epsilon L^{n-1}$, where

$L = \max\{(b_j - a_j) \mid 1 \leq j \leq n\}$.

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Question 23

$$\int_1^{\infty} \frac{\log x + \sin x}{\sqrt{x}}\, dx$$

Ans: **Ans**

$\frac{\log x + \sin x}{\sqrt{x}} > \frac{1}{\sqrt{x}}$ if $x > e^2$. Since $\int_{e^2}^{\infty} \frac{dx}{\sqrt{x}} = \infty$, $\int_{e^2}^{\infty} \frac{\log x + \sin x}{\sqrt{x}}\, dx = \infty$.
Therefore, $\int_1^{\infty} \frac{\log x + \sin x}{\sqrt{x}}\, dx = \infty$

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Question 24

$$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1 + x^2}}\, dx$$

Ans:

$\frac{1 + \cos^2 x}{\sqrt{1 + x^2}} > \frac{1}{\sqrt{1 + x^2}} = \frac{1}{x\sqrt{1 + 1/x^2}} > \frac{1}{x\sqrt{2}}$ if $x > 1$. Since $\int_1^{\infty} \frac{dx}{x} = \infty$,
$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1 + x^2}}\, dx = \infty$

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Question 25

$$\int_0^\infty \frac{4+\cos x}{(1+x)\sqrt{x}}\,dx$$

Ans:

$I = I_1 + I_2$ where $I_1 = \int_0^1 \frac{4+\cos x}{(1+x)\sqrt{x}}dx$ and $I_2 = \int_1^\infty \frac{4+\cos x}{(1+x)\sqrt{x}}dx$. $0 < \frac{4+\cos x}{(1+x)\sqrt{x}} < \frac{5}{\sqrt{x}}$ if $x > 0$. Since $\int_0^1 \frac{dx}{\sqrt{x}}$, $I_1 < \infty$. $0 < \frac{4+\cos x}{(1+x)\sqrt{x}} < \frac{5}{x^{3/2}}$ if $x > 1$. Since $\int_0^1 \frac{dx}{x^{3/2}}$, $I_2 < \infty$ Therefore, I converges.

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Question 26

The function $f(x,y) = \frac{xy}{x^2+y^2}$ is defined everywhere in $\mathbb{R}^2$ except at $(0,0)$.

Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

Ans:

by letting (x,y) approach $(0,0)$ along the line $y = x$, we see the functional values

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}$$ and conclude that the limit is $1/2$.

However, if we let (x,y) approach $(0,0)$ along the line $y = -x$, we see the functional values

$$f(x,-x) = -\frac{x^2}{2x^2} = -\frac{1}{2}$$

and conclude that the limit equals $-1/2$.

In fact, they are both incorrect. What we have shown is that

$$\lim_{x\to 0} f(x,x) = \frac{1}{2} \quad \text{and} \quad \lim_{x\to 0} f(x,-x) = -\frac{1}{2}.$$

Since $\lim_{x\to 0} f(x,x)$ and $\lim_{x\to 0} f(x,-x)$ must both equal $\lim_{(x,y)\to(0,0)} f(x,y)$.

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Question 27

If

$$f(x,y,z) = \cos\left(\frac{1}{x^2+2y^2+z^2}\right),$$

then

$$\lim_{|\mathbf{X}|\to\infty} f(\mathbf{X}) = 1.$$

Ans:

the continuity of $\cos u$ at $u = 0$ implies that f $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|\cos u - 1| < \varepsilon \quad \text{if} \quad |u| < \delta.$$

Since

$$\frac{1}{x^2+2y^2+z^2} \le \frac{1}{|\mathbf{X}|^2}.$$

It follows that if $|\mathbf{X}| > 1/\sqrt{\delta}$, then

$$\frac{1}{x^2+2y^2+z^2} < \delta.$$

Therefore,

$$|f(\mathbf{X}) - 1| < \varepsilon.$$

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Question 28

Suppose that $\mathbf{U}_0$ is in T and is a limit point of T, $\mathbf{G}$ is continuous at $\mathbf{U}_0$, and f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$. Then $h = f \circ \mathbf{G}$ is continuous at $\mathbf{U}_0$.

Proof: Suppose that $\varepsilon > 0$. Since f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$, there is an $\varepsilon_1 > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{G}(\mathbf{U}_0))| < \varepsilon \qquad (1)$$

if

$$|\mathbf{X} - \mathbf{G}(\mathbf{U}_0)| < \varepsilon_1 \quad \text{and} \quad \mathbf{X} \in D_f. \qquad (2)$$

Since $\mathbf{G}$ is continuous at $\mathbf{U}_0$, there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)| < \varepsilon_1 \quad \text{if} \quad |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

By taking $\mathbf{X} = \mathbf{G}(\mathbf{U})$ in (1) and (2)

$$|h(\mathbf{U}) - h(\mathbf{U}_0)| = |f(\mathbf{G}(\mathbf{U}) - f(\mathbf{G}(\mathbf{U}_0))| < \varepsilon$$

if

$$|\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in T.$$

$$\sqrt{x^2+2y^2+z^2} < o.$$

Therefore,

$$|f(\mathbf{X}) - 1| < \varepsilon.$$

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Question 29

Linear Function

A *linear function* is a function of the form $L(\mathbf{X}) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$,

where $m_1, m_2, \ldots, m_n$ are constants. From definition of differentiability, f is differentiable at $\mathbf{X}_0$ if and only if there is a linear function L such that $f(\mathbf{X}) - f(\mathbf{X}_0)$ can be approximated so well near $\mathbf{X}_0$ by

$$L(\mathbf{X}) - L(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0)$$

that

$$f(\mathbf{X}) - f(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})(|\mathbf{X} - \mathbf{X}_0|),$$

where

$$\lim_{\mathbf{X}\to\mathbf{X}_0} E(\mathbf{X}) = 0.$$

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Question 30

If

$$f(x,y) = \frac{x^2+y^2}{x-y},$$

Solution:

$$f_x(x,y) = \frac{2x}{x-y} - \frac{x^2+y^2}{(x-y)^2}$$

$$f_y(x,y) = \frac{2y}{x-y} + \frac{x^2+y^2}{(x-y)^2}.$$

Since f_x and f_y are continuous on

$$S = \{(x,y) : x \neq y\}$$

f is continuously differentiable on S.

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Question 31

Absolute integrability

We say that f is *absolutely integrable on* $[a,b)$ if f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|\,dx < \infty$. In this case we also say that $\int_a^b f(x)\,dx$ *converges absolutely* or *is absolutely convergent*.

Conditional convergence

An oscillatory function may be integrable, but not absolutely integrable, on $[a,b)$, as the next example shows. We then say that f is *conditionally* integrable on $[a,b)$, and that $\int_a^b f(x)\,dx$ converges *conditionally*.

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Question 32

Dirichlet's Test

Suppose that f is continuous and its antiderivative $F(x) = \int_a^x f(t)\,dt$ is bounded on $[a,b)$.

Let g' be absolutely integrable on $[a,b)$, and suppose that

$\lim_{x\to b-} g(x) = 0$. Then $\int_a^b f(x)g(x)\,dx$ converges.

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Differentiable Surfaces

A *differentiable surface* S in $\mathbb{R}^n$ $(n > 1)$ is the image of a compact subset D of $\mathbb{R}^m$, where $m < n$, under a continuously differentiable transformation $\mathbf{G} : \mathbb{R}^m \to \mathbb{R}^n$. If $m = 1$, S is also called a *differentiable curve*.

Question 33

$$\int_1^\infty x^q \sin x\,dx$$

diverges if $q > 0$, but none of the other tests that we have studied so far implies this. It is not enough to argue that the integrand does not approach zero as $x \to \infty$ (a common mistake), since this does not imply divergence. To see that the integral diverges, we observe that if it converged for some $q > 0$, then $F(x) = \int_1^x x^q \sin x\,dx$ would be bounded on $[1,\infty)$.

We could let

$$f(x) = x^q \sin x \quad \text{and} \quad g(x) = x^{-q}$$

in Dirichlet's test and conclude that

$$\int_1^\infty \sin x\,dx$$

also converges. This is false.

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Question 34

Upper Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is a partition of R.

Let

$$M_j = \sup_{\mathbf{X}\in R_j} f(\mathbf{X}).$$

The *upper sum* of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^{k} M_j V(R_j).$$

The *upper integral of* f *over* R, denoted by $\overline{\int_R} f(\mathbf{X})\,d\mathbf{X}$,

is the infimum of all upper sums.

Lower Integrals:

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is a partition of R.

$$m_j = \inf_{\mathbf{X}\in R_j} f(\mathbf{X}).$$

The *lower sum of* f *over* $\mathbf{P}$ is

$$s(\mathbf{P}) = \sum_{j=1}^{k} m_j V(R_j).$$

The *lower integral of* f *over* R, denoted by $\underline{\int_R} f(\mathbf{X})\,d\mathbf{X}$,

is the supremum of all lower sums.

Question 35

Any bounded set E with only finitely many limit points has zero content.

To see this, we first observe that if E has no limit points, then it must be finite, by the Bolzano–Weierstrass theorem, and therefore must have zero content.

Now suppose that the limit points of E are $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m$. Let $R_1, R_2, \ldots, R_m$ be rectangles such that $\mathbf{X}_i \in R_i^0$ and

$$V(R_i) < \frac{\varepsilon}{2m}, \quad 1 \le i \le m. \tag{3.30}$$

The set of points of E that are not in $\cup_{j=1}^m R_j$ has no limit points (why?) and, being bounded, must be finite (again by the Bolzano–Weierstrass theorem).

If this set contains p points, then it can be covered by rectangles $R_1', R_2', \ldots, R_p'$ with

$$V(R_j') < \frac{\varepsilon}{2p}, \quad 1 \le j \le p. \tag{3.31}$$

Now,

$$E \subset \left(\bigcup_{i=1}^m R_i\right) \bigcup \left(\bigcup_{j=1}^p R_j'\right)$$

From (3.30) and (3.31),

$$\sum_{i=1}^m V(R_i) + \sum_{j=1}^p V(R_j') < \varepsilon.$$

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Question 36

Example: The circle

$\{(x,y) : x^2 + y^2 = 9\}$ is a differentiable curve in $\mathbb{R}^2$.

Ans:

Since it is the image of $D = [0, 2\pi]$ under the continuously differentiable transformation $\mathbf{G} : \mathbb{R} \to \mathbb{R}^2$ defined by

$$\mathbf{X} = \mathbf{G}(\theta) = \begin{bmatrix} 3\cos\theta \\ 3\sin\theta \end{bmatrix}.$$

Example: The sphere

$\{(x,y,z) : x^2 + y^2 + z^2 = 4\}$ is a differentiable surface in $\mathbb{R}^3$.

Ans:

Since it is the image of $D = \{(\theta,\phi) : 0 \le \theta \le 2\pi, -\pi/2 \le \phi \le \pi/2\}$

under the continuously differentiable transformation $\mathbf{G} : \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$\mathbf{X} = \mathbf{G}(\theta,\phi) = \begin{bmatrix} 2\cos\theta\cos\phi \\ 2\sin\theta\cos\phi \\ 2\sin\phi \end{bmatrix}.$$

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Question 37

Iterated Integrals I_1

assume that f is continuous on $R = [a,b] \times [c,d]$

Then, for each y in $[c,d]$, $f(x,y)$ is continuous with respect to x on $[a,b]$, so the integral

$$F(y) = \int_a^b f(x,y)\,dx$$

exists.

Moreover, the uniform continuity of f on R implies that F is continuous and therefore integrable on $[c,d]$.

We say that

$$I_1 = \int_c^d F(y)\,dy = \int_c^d \left(\int_a^b f(x,y)\,dx \right) dy$$

is an *iterated integral* of f over R.

which we usually write as

$$I_1 = \int_c^d dy \int_a^b f(x,y)\,dx.$$

Iterated Integrals I_2

Another iterated integral can be defined by writing

$$G(x) = \int_c^d f(x,y)\,dy, \quad a \le x \le b,$$

We say that

$$I_2 = \int_a^b G(x)\,dx = \int_a^b \left(\int_c^d f(x,y)\,dy \right) dx,$$

which we usually write as

$$I_2 = \int_a^b dx \int_c^d f(x,y)\,dy.$$

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Question 38

The function $f(x) = (1-x)^{-p}$ is locally integrable on $[0,1)$ and, if $p \neq 1$ and $0 < c < 1$,

$$\int_0^c (1-x)^{-p}\,dx = \frac{(1-x)^{-p+1}}{p-1}\bigg|_0^c = \frac{(1-c)^{-p+1}-1}{p-1}.$$

Hence,

$$\lim_{c\to 1-} \int_0^c (1-x)^{-p}\,dx = \begin{cases} (1-p)^{-1}, & p < 1, \\ \infty, & p > 1. \end{cases}$$

For $p = 1$,

$$\lim_{c\to 1-} \int_0^c (1-x)^{-1}\,dx = -\lim_{c\to 1-} \log(1-c) = \infty.$$

Hence,

$$\int_0^1 (1-x)^{-p}\,dx = \begin{cases} (1-p)^{-1}, & p < 1, \\ \infty, & p \ge 1. \end{cases}$$

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Question 39

The function $f(x) = x^{-p}$ is locally integrable on $[1,\infty)$ and, if $p \neq 1$ and $c > 1$,

$$\int_1^c x^{-p}\,dx = \frac{x^{-p+1}}{-p+1}\bigg|_1^c = \frac{c^{-p+1}-1}{-p+1}.$$

Hence,

$$\lim_{c\to\infty} \int_1^c x^{-p}\,dx = \begin{cases} (p-1)^{-1}, & p > 1, \\ \infty, & p < 1. \end{cases}$$

For $p = 1$,

$$\lim_{c\to\infty} \int_1^c x^{-1}\,dx = \lim_{c\to\infty} \log c = \infty.$$

Hence,

$$\int_1^\infty x^{-p}\,dx = \begin{cases} (p-1)^{-1}, & p > 1, \\ \infty, & p \le 1. \end{cases}$$

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Question 40

The function $f(x) = \cos x$ is locally integrable on $[0,\infty)$

$$\lim_{c\to\infty} \int_0^c \cos x\,dx = \lim_{c\to\infty} \sin c$$

does not exist; thus, $\int_0^\infty \cos x\,dx$ diverges, but not to $\pm\infty$.

Question

The function $f(x) = \log x$ is locally integrable on $(0,1]$, but unbounded as $x \to 0+$. Since

$$\lim_{c\to 0+} \int_c^1 \log x\,dx = \lim_{c\to 0+} (x\log x - x)\Big|_c^1 = -1 - \lim_{c\to 0+} (c\log c - c) = -1,$$

$$\int_0^1 \log x\,dx = -1.$$

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Question 41

Let $f(x,y) = x + y$ and $R = [0,1] \times [1,2]$.

Ans:

$$F(y) = \int_0^1 f(x,y)\,dx = \int_0^1 (x+y)\,dx = \frac{1}{2} + y$$

$$I_1 = \int_1^2 F(y)\,dy = \int_1^2 \left(\frac{1}{2} + y\right) dy = \left(\frac{y}{2} + \frac{y^2}{2}\right)\Big|_1^2 = 2.$$

Also,

$$G(x) = \int_1^2 (x+y)\,dy = \left(xy + \frac{y^2}{2}\right)\Big|_{y=1}^2 = x + \frac{3}{2},$$

$$I_2 = \int_0^1 G(x)\,dx = \int_0^1 \left(x + \frac{3}{2}\right) dx = \left(\frac{x^2}{2} + \frac{3x}{2}\right)\Big|_0^1 = 2.$$

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Question 42

Theorem: Suppose that f is integrable on $R = [a,b] \times [c,d]$ and $F(y) = \int_a^b f(x,y)\,dx$ exists for each y in $[c,d]$.

Ans:

Then F is integrable on $[c,d]$, and

$$\int_c^d F(y)\,dy = \int_R f(x,y)\,d(x,y);$$

that is,

$$\int_c^d dy \int_a^b f(x,y)\,dx = \int_R f(x,y)\,d(x,y).$$

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